

Mahmoud Mohamed Hassanen

Data Engineer | Big Data Engineer | ETL Engineer

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Career Objective:

Highly motivated and self-driven Data Engineer with a strong foundation in Python, SQL, and cloud technologies. Eager to leverage my expertise in data manipulation, model building, and cloud computing to solve complex problems and contribute to the success of a dynamic team.

Education:

Bachelor's degree from Tanta University, Faculty of Engineering, Computer Engineering and Automatic Control. (2018 - 2023)

Graduation Project: Website for Semantic Search in Videos based on Deep Learning Using SBERT (Grade: Excellent.) %

Role: Managed the data pipeline, including data extraction, transformation, and formatting to suit the model.

Training Diploma:

Data Engineering | Intensive Code Camp (May 2024 – September 2024)

(ITI) Information Technology Institute, (MCIT) Ministry of Communications and Information Technology, Menoufia branch.

Internships:

- **NLP R&D Intern Engineer at smsmAI** (March 2024 – June 2024)
Gained hands-on experience in real-world applications.
 1. RAG Using Langchain and open-source Embedding Model
 2. Fine Tuning Translation NLLB Model on Custom Data (EN-AR)
- **Data Science Intern at Banque Misr, New Cairo** (July 2023 - August 2023)
Conducted data preprocessing, model building, and presented findings to stakeholders.
Gained hands-on experience in real-world data science applications.
 1. Loan Defaulter Predictor Classifier.
 2. ATM Location Optimization.

Projects:

- **Churn Customer End-to-End Pipeline**
Developed a data pipeline for churn prediction, Using PySpark processed the data in three branches:
 1. Ingested data using kafka and airflow and Stored raw data in Azure SQL DB.
 2. Ran classification models with PySpark ML, saving results to SQL Server and visualized in Power BI.
 3. Streamed data to Hadoop with Hive SQL for querying.
- **Gravity Books Data Warehouse ETL (SSIS)**
Designed and implemented ETL processes using SSIS to move data from a SQL Server database to a SQL Server Data Warehouse.
Created and optimized data warehouse tables to support efficient querying and reporting.
- **Global Terrorism Dataset Analysis**
Executed comprehensive data analysis on global terrorism, utilizing Python, Pandas, and visualization tools to reveal trends and insights.
- **BERT Modeling – Bertopic**
Developed topic models with Bertopic, leveraging BERT to categorize and analyze textual data.

Technical Skills:

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|-----------|----------------------|-----------------------------------|
| • Python | • ETL (Talend, SSIS) | • Cloud Fundamentals (AWS, Azure) |
| • SQL | • NoSQL (MongoDB) | • Data Warehouse concepts |
| • Hadoop | • Database Design | • HuggingFace |
| • PySpark | • CI/CD Jenkins | • Linux |
| • Airflow | • Docker | • Kafka (Essential) |

Courses:

- **AWS Cloud Foundation | AWS Academy**
Practical knowledge for IAM, S3, EC2, Lambda.
- **Associate Data Engineer in SQL | Datacamp**
Python, SQL, and Git Fundamentals
Hands-on data manipulation and relational database querying

Proficiency in data cleaning, code optimization, and testing

- **Generative AI | Udemy**
Foundations of Generative models
Building NLP applications using OpenAI API and Hugging Face Transformers

Extracurriculars:

- **Public Relations and Fundraiser at IEEE TSB**, Tanta (May 2021 – 2023 August)
Gathered contacts for speakers and sponsors.
Communicated with speakers and sponsors.
- **Public Relations member at LifeMakers**, Tanta (October 2020 - May 2023)
Raised funds for the organization.
Participated in various events.

Languages Skills:

- English (V. Good)
- Arabic (Native)

Personal Information:

- Military: Exemption